Lecture 14 : Project Evaluation and Programme Management (Contd.)

Introduction

In the dynamic landscape of project management, making informed decisions is paramount to success. This lecture delves deeper into the critical final stages of Cost-Benefit Analysis (CBA) and introduces robust methodologies for evaluating project risks. Understanding how to interpret the outcomes of a CBA and subsequently take appropriate action is crucial for project managers. Beyond the initial financial assessments, projects are inherently exposed to various uncertainties, necessitating a systematic approach to risk evaluation.

This session will equip students with the knowledge to not only finalize their project evaluations but also to proactively identify, quantify, and manage the inherent risks. By exploring techniques such as risk matrices, adjustments to Net Present Value (NPV) calculations, and the strategic use of decision trees, we aim to provide a comprehensive framework for navigating the complexities of project selection and execution. The ability to assess and mitigate risks effectively can often be the deciding factor between a project's triumph and its failure, making these concepts indispensable for any aspiring project manager.

Core Concepts

Term	Definition	Significance
`Cost-Benefit Analysis (CBA)`	A systematic process for calculating and comparing the benefits and costs of a project or decision, used to determine if it is a sound investment.	It provides a structured way to evaluate project viability by weighing financial and non-financial outcomes.
`Valuation Problems (CBA)`	Difficulties in identifying, measuring, and quantifying intangible costs and benefits within a cost-benefit analysis.	These challenges can lead to an incomplete or inaccurate assessment of a project's true value.
`Distortion Problems (CBA)`	Issues arising from intentional favoritism for political reasons	They compromise the objectivity and reliability of the analysis, potentially

Term	Definition	Significance
	or incomplete/missing data, leading to skewed CBA results.	leading to suboptimal project choices.
`Project Risk`	Threats related to the successful execution and completion of a project, impacting its schedule, budget, or quality.	Managing these risks is vital for ensuring the project is delivered as planned.
`Business Risk`	Factors that threaten the expected benefits or outcomes of a delivered project, impacting its long-term value or profitability.	This type of risk is a primary focus in business case development, as it directly affects the strategic goals.
`Project Risk Matrix`	A tool used to identify and classify potential project risks based on their relative importance (impact) and likelihood (probability of occurrence).	It helps prioritize risks, allowing project managers to focus mitigation efforts on the most critical threats.
`Risk Premium (NPV)`	An additional percentage added to the discount rate when calculating Net Present Value (NPV) for projects deemed relatively risky.	It accounts for the increased uncertainty, making riskier projects appear less attractive unless their returns are significantly higher.
`Decision Trees`	A graphical representation of possible outcomes of a series of related choices, used to evaluate risks and potential future options.	They enable project managers to visualize complex decision paths, probabilities, and expected values to make strategic choices.
`Sensitivity Analysis`	A technique used in risk profile analysis to determine how different values of an	It helps identify which factors are most critical to a project's success, guiding

Term	Definition	Significance
	independent variable impact a particular dependent variable.	where to focus control or mitigation efforts.

Detailed Analysis

1. Completing the Cost-Benefit Analysis (CBA)

The final two steps of a comprehensive Cost-Benefit Analysis involve interpreting the results and taking appropriate action. After the initial evaluation techniques have been applied, the project's outcomes must be thoroughly understood.

- **Interpreting the Results**:
- This phase requires comparing the project's actual results against a predefined standard or the outcomes of alternative investment opportunities.
- Interpretation is inherently subjective, relying heavily on the project manager's judgment and intuition.
- Project managers may encounter either a single, known value for measures like Net Present Value (NPV) or a range of possible values, depending on the level of uncertainty.
- For simpler scenarios (single or range of values), techniques like net profit analysis might be used due to their ease of implementation. However, it's noted that net profit analysis typically doesn't account for the time value of money, a limitation that, if addressed, could make it comparable to NPV.
- **Taking Appropriate Action**:
- Following interpretation, the project manager must make a decision or take action, which is also highly subjective.
- This decision is influenced by the project manager's or end-user's confidence in the estimated costs and benefits, as well as the overall magnitude of the investment.
- The ultimate goal is to select the most cost-effective and beneficial system for the user.
- A crucial output of this stage is the preparation of a feasibility study report, which documents the major findings from the analysis and provides suggested recommendations.

2. Limitations of Cost-Benefit Analysis

Despite its utility, CBA is not without its drawbacks, which can impact the accuracy and reliability of its conclusions.

- **Valuation Problems**:
- A significant challenge lies in identifying, measuring, and quantifying intangible costs and benefits. Items like improved morale, enhanced reputation, or reduced stress are difficult to assign a monetary value, leading to potential underestimation or omission.
- **Distortion Problems**:
- Results can be distorted in two primary ways:
- **Intentional Favoritism**: Political motivations can lead to biased assessments, favoring a particular alternative regardless of its true merit.
- **Incomplete or Missing Data**: When critical data is unavailable or overlooked, the analysis becomes flawed, making it difficult to take appropriate action.
- **Completeness Problems**:
- The estimated costs for a CBA might be inflated (on the high side) or, conversely, insufficient costs might be considered, leading to an incomplete analysis. This can misrepresent the true financial picture of the project.

3. Dealing with Uncertainty: Risk Evaluation

Every project carries inherent risks, which can be broadly categorized into two types:

- **Project Risks**: These are threats directly related to the successful execution of the project itself, potentially impacting its timeline, budget, or quality.
- **Business Risks**: These risks pertain to factors that threaten the expected benefits or the overall value of the delivered project. The business case for a project primarily focuses on evaluating these business risks.

4. Risk Identification and Ranking

A foundational step in risk management is to identify potential risks and quantify their potential effects.

- **Project Risk Matrix**: One effective approach is to construct a project risk matrix. This matrix utilizes a checklist of possible risks and then classifies them based on two key dimensions:
- **Importance**: How significant would the impact be if the risk occurred (e.g., High, Medium, Low)?
- **Likelihood**: What is the probability of the risk occurring (e.g., High, Medium, Low)?
- **Separate Assessment**: It is crucial to assess importance and likelihood separately. A very serious risk with a low probability might be treated differently from a less serious risk with a very high probability of occurrence. For instance, a project with a better return but higher risk (Project A) needs careful consideration compared to a project with lower return but less risk (Project B).

5. Risk and Net Present Value (NPV)

When dealing with projects of varying risk levels, the Net Present Value calculation can be adjusted to reflect this uncertainty.

- **Higher Discount Rate**: For relatively risky projects, it is common practice to use a higher discount rate when calculating NPV. This higher rate acts as a "risk premium."
- **Risk Premium Application**: For example, a reasonably safe project might incur an additional 2% risk premium on its discount rate, while a fairly risky project might warrant an additional 5%. If the normal discount rate is 10%, a safe project might use 12%, and a risky one 15%.
- **Categorization**: Projects can be categorized (e.g., high, medium, low risk) using a scoring method, with specific risk premiums designated for each category. Even if these premiums are somewhat arbitrary, they provide a consistent method for incorporating risk into financial evaluations.

6. Cost-Benefit Analysis Method for Handling Risks

A more sophisticated approach integrates probabilistic outcomes into the CBA framework.

- **Probabilistic Outcomes**: Instead of a single cash flow forecast, this method considers multiple possible outcomes, each with an estimated probability of occurring and a corresponding value.
- **Weighted Sum**: The project's value is then determined by summing the cost or benefit for each possible outcome, weighted by its associated probability.
- **Example: Payroll Application**:
- A company developing a payroll application faces three scenarios:
- High sales ($800,000 annual income) with 10% probability.
- Medium sales ($650,000 annual income) with 60% probability.
- Low sales ($100,000 annual income) due to competition, with 30% probability.
- The expected annual income is calculated as:
  (800,000 * 0.1) + (650,000 * 0.6) + (100,000 * 0.3) = 80,000 + 390,000 + 30,000 = $500,000.
- Given development costs of $750,000, sales constant for 4 years, and annual maintenance costs of $200,000:
- Total expected income over 4 years = ($500,000 * 4) - ($200,000 * 4) = $2,000,000 - $800,000 = $1,200,000.
- Comparing this $1,200,000 benefit to the $750,000 investment suggests a good return.
- **Crucial Consideration**: However, the worst-case scenario (low sales) has a 30% chance of occurring. If this happens, the company would lose money, making the project too risky despite the average positive return. Therefore, it is not advisable to proceed.
- **Limitations**:
- **Probability Assignment**: Accurately assigning probabilities to various outcomes is a significant challenge.
- **Worst-Case Scenarios**: This method often focuses on average or expected outcomes and may not fully account for the severe impact of worst-case scenarios.

7. Risk Profile Analysis

This approach aims to overcome some of the limitations of the probabilistic CBA by constructing a risk profile using sensitivity analysis.

- **Sensitivity Analysis**: By studying the results of sensitivity analysis, project managers can identify which factors are most critical to the project's success.
- **Control and Mitigation**: Once critical factors are identified, the focus shifts to deciding whether better control can be exercised over them or if their effects can be mitigated.
- **Decision Point**: If neither control nor mitigation is feasible, the project manager must either accept the inherent risks or abandon the project altogether.

8. Using Decision Trees to Evaluate Risks

Decision trees offer a powerful visual and analytical tool for evaluating risks, especially when decisions affect future options.

- **Beyond Passive Observation**: Unlike previous methods that might treat project managers as passive observers of risk, decision trees enable active decision-making in the face of uncertainty.
- **Future Options**: Decisions made today can significantly impact future project profitability. Decision trees help assess how current choices will affect these future outcomes.
- **Example: System Replacement**:
- A company considers replacing or extending its sales order processing system.
- **Extending the system**:
- Market expands (20% probability): NPV of -$100,000 (loss due to lost revenue).
- Market does not expand (80% probability): NPV of $75,000.
- Expected value = (75,000 * 0.8) + (-100,000 * 0.2) = 60,000 - 20,000 = $40,000.
- **Replacing the system**:
- Market expands (20% probability): NPV of $250,000.
- Market does not expand (80% probability): NPV of -$50,000.
- Expected value = (250,000 * 0.2) + (-50,000 * 0.8) = 50,000 - 40,000 = $10,000.

- **Conclusion**: Comparing the expected values, extending the existing system ($40,000) yields a higher expected benefit than replacing it ($10,000). Therefore, the company should choose to extend the existing system.
- **Analysis**: Decision trees involve evaluating the expected benefit of each path from a decision point by summing the values of possible outcomes multiplied by their probabilities.

Key Takeaways

- [ ] Cost-Benefit Analysis (CBA) concludes with interpreting results and taking appropriate action, both of which are subjective and require judgment.
- [ ] Feasibility study reports are crucial outputs, summarizing findings and recommendations after CBA.
- [ ] Limitations of CBA include difficulties in valuing intangibles, potential for data distortion, and issues with cost completeness.
- [ ] Project risks relate to execution, while business risks threaten delivered benefits; the latter is key for business cases.
- [ ] Risk identification and ranking use a project risk matrix to classify risks by importance and likelihood.
- [ ] Higher discount rates (risk premiums) are applied in NPV calculations for riskier projects to account for uncertainty.
- [ ] Probabilistic CBA considers multiple outcomes with associated probabilities, but assigning these probabilities and accounting for worst-case scenarios remain challenges.
- [ ] Risk profile analysis uses sensitivity analysis to identify critical success factors and guide mitigation or control efforts.
- [ ] Decision trees are powerful tools for evaluating risks and future options by calculating expected values for different decision paths.
- [ ] A project's success hinges not just on good management but also on sound initial evaluation and proactive risk management.

Conclusion

This lecture has underscored the critical importance of moving beyond initial project evaluations to a comprehensive understanding of both the financial implications and inherent risks. By meticulously interpreting the results of a Cost-Benefit Analysis and taking decisive action, project managers lay the groundwork for successful project implementation. However, the journey doesn't end there; the ability to identify, quantify, and strategically manage risks is what truly differentiates robust project planning.

From employing project risk matrices to systematically categorize threats, to adjusting financial models like NPV with risk premiums, and leveraging the analytical power of decision trees, the tools discussed provide a holistic framework. These methodologies empower project managers to make informed choices, not merely reacting to uncertainties but proactively shaping a project's trajectory. Ultimately, the integration of thorough evaluation and astute risk management ensures that projects are not only viable but also resilient, capable of delivering sustained value in an unpredictable environment.